

Digital Signal Processing

Digital Filters II

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FIR Filter

Finite, non recursive filter
 $H(z) = B(z)$

$$H(z) = 3 + 4z^{-1} + 3z^{-2}$$

Coefficients (taps) coincide
with impulse response

$$h[n] = 3\delta[n] + 4\delta[n-1] + 3\delta[n-2]$$

3 coeffs $\Rightarrow$
2nd order

Many algorithms are used
for finding coefficients

```
1 f=[0 0.25 0.3 0.55 0.6 0.85 0.9 1]; %frequencies
2 A=[1 1 0 0 1 1 0 0]; %amplitudes
3 M=20; %filter order
4 W=[1000 1 1 1]; %weights, first passband is prioritized
5 B=firls(M,f,A,W);
6 freqz(B);
```

firls uses Least Squares

FIR Filter – Windowing (LP filter)

1. Find ideal filter

$$h[n] = \frac{\sin(0.3n)}{\pi n}, n \in [\dots, -1, 0, 1, \dots]$$

2. Window of order M (assume M is even)

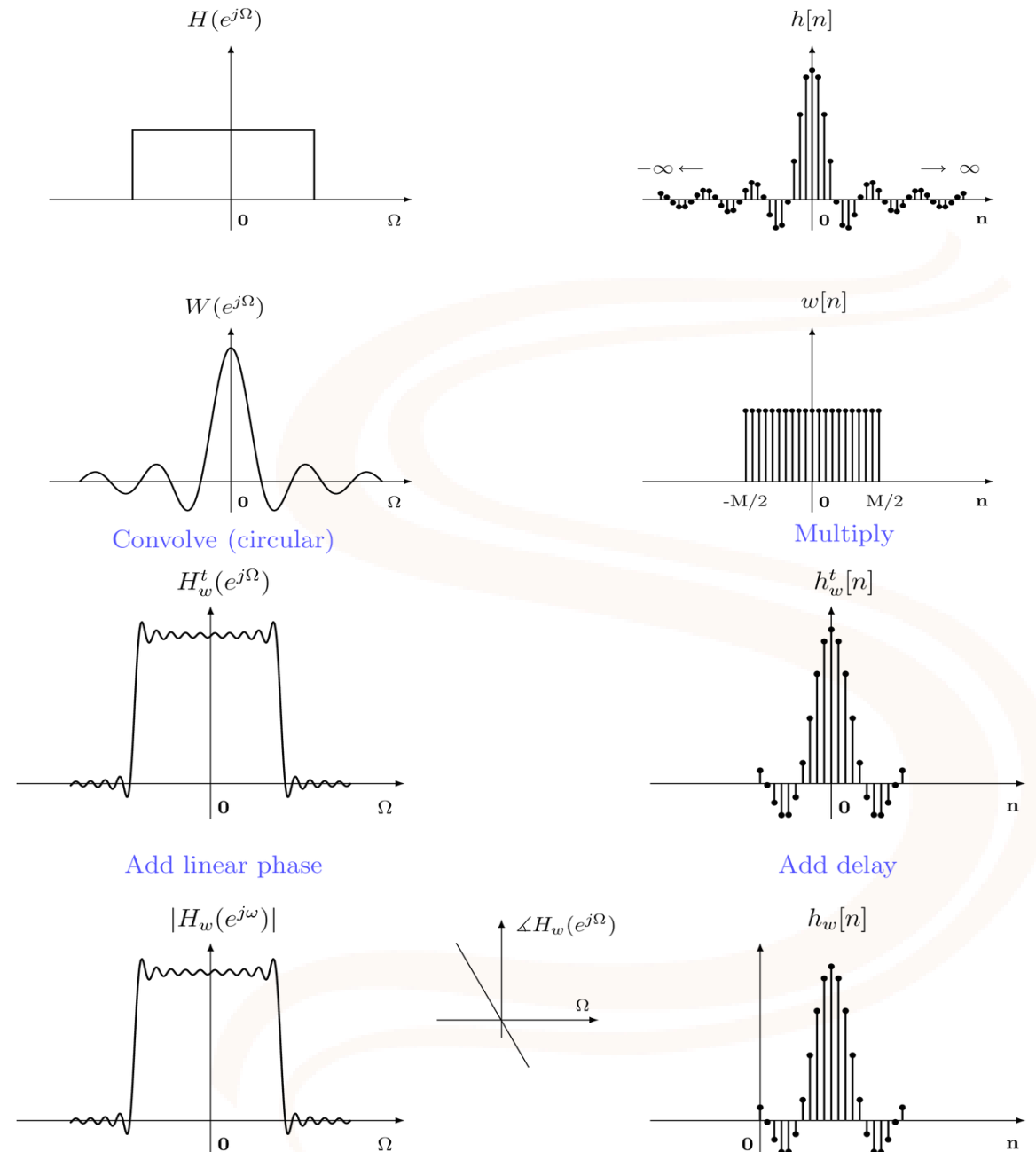
$$w[n] = 1, n \in [-4, -3, \dots, 4]$$

3. Multiply $h[n]$ and $w[n]$

$$h_w^t[n] = h[n], n \in [-4, -3, \dots, 4]$$

4. Delay by M/2 to make a causal filter

$$h_w[n] = \frac{\sin(0.3(n-4))}{\pi(n-4)}, n \in [0, 1, \dots, 8]$$



FIR Filter – DC gain

DC gain is obtained at $z = e^{j0} = 1$

$$H_1(z) = 1 + 2z^{-1} + 2z^{-2} + z^{-3}$$

$$H_1(1) = 1 + 2 + 2 + 1 = 6$$

$$H_2(z) = 3 - z^{-1} - 1z^{-2} + 3z^{-3}$$

$$H_2(1) = 3 - 1 - 1 + 3 = 4$$

$$H_3(z) = 1 + 2z^{-1} - 2z^{-2} - z^{-3}$$

$$H_3(1) = 1 + 2 - 2 - 1 = 0$$

Unitary DC gain When $H(1) = 1$

$$\begin{aligned} H'_1(z) &= H_1(z)/6 \\ H'_2(z) &= H_2(z)/4 \end{aligned}$$

$H'_3(z)$ cannot have unitary DC gain

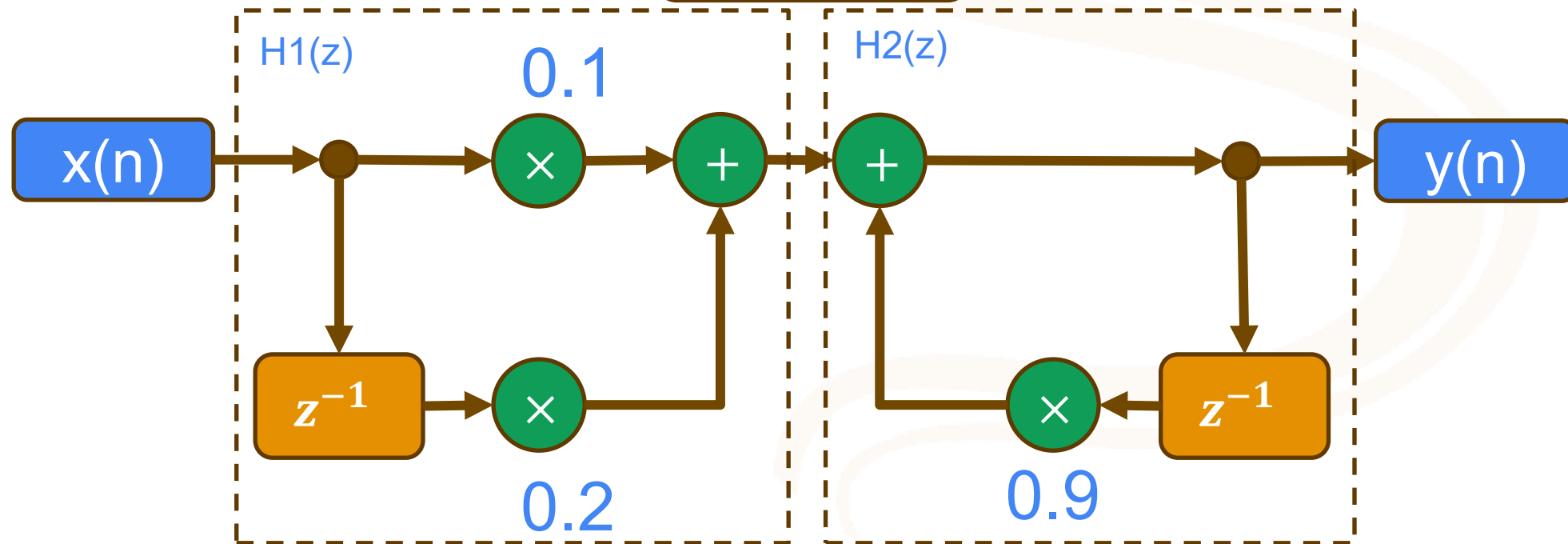
Digital filter realizations – Direct Form I

$$H(z) = \frac{0.1 + 0.2z^{-1}}{1 - 0.9z^{-1}}$$

$$\frac{y[n] - 0.9y[n-1]}{0.1x[n] + 0.2x[n-1]}$$

$$y[n] = 0.1x[n] + 0.2x[n-1] + 0.9y[n-1]$$

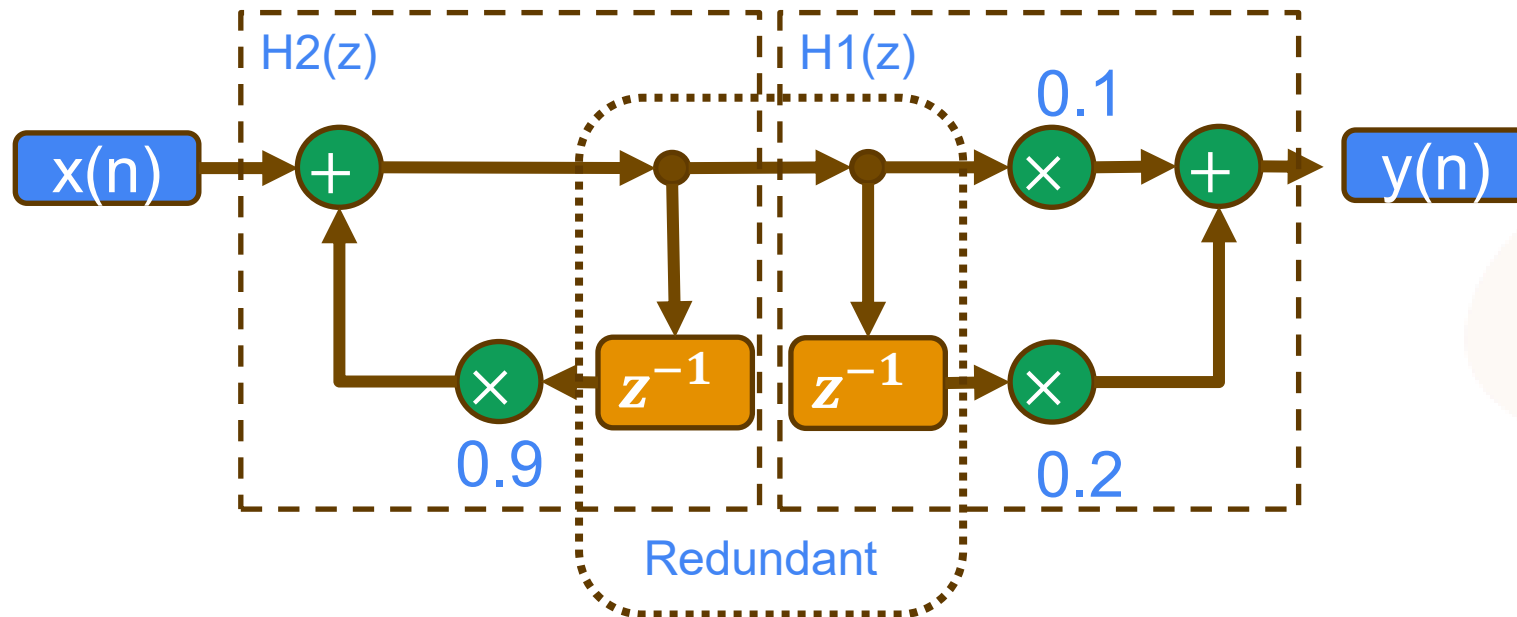
Zeros First
Poles Last



Two memory
cells

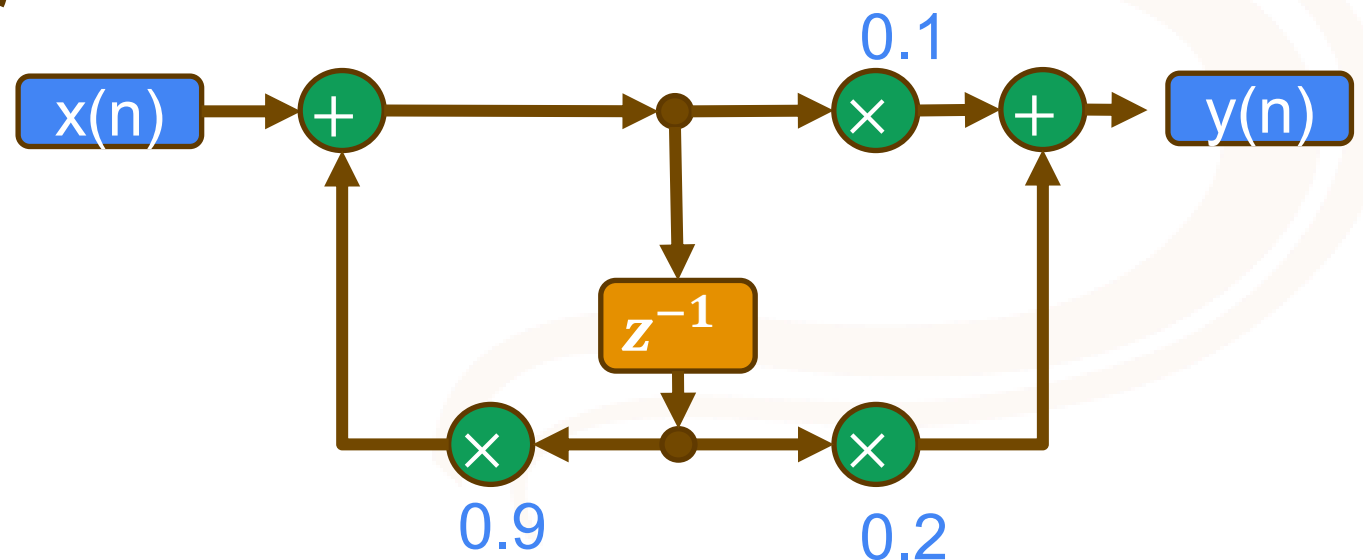
Digital filter realizations – Direct Form II

$$H(z) = \frac{0.1 + 0.2z^{-1}}{1 - 0.9z^{-1}}$$



Poles First
Zeros Last

Merge the
delay blocks



Digital filter realizations – Transposed Direct Form II

$$H(z) = \frac{0.1 + 0.2z^{-1}}{1 - 0.9z^{-1}}$$

Switch $x[n]$ and $y[n]$ positions

Adders become dots
Dots become adders

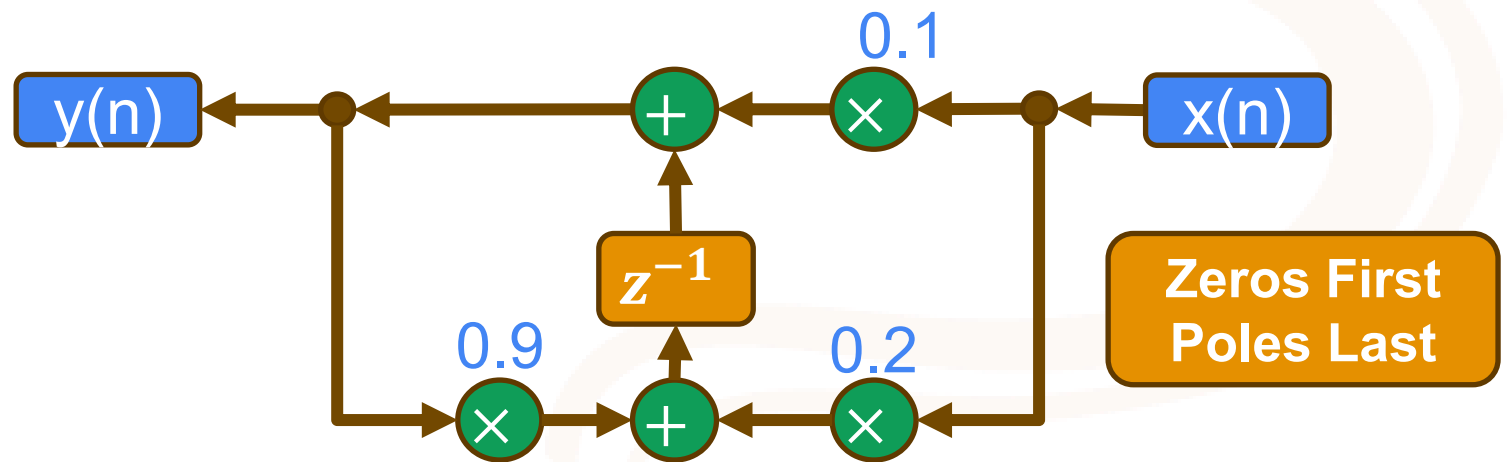
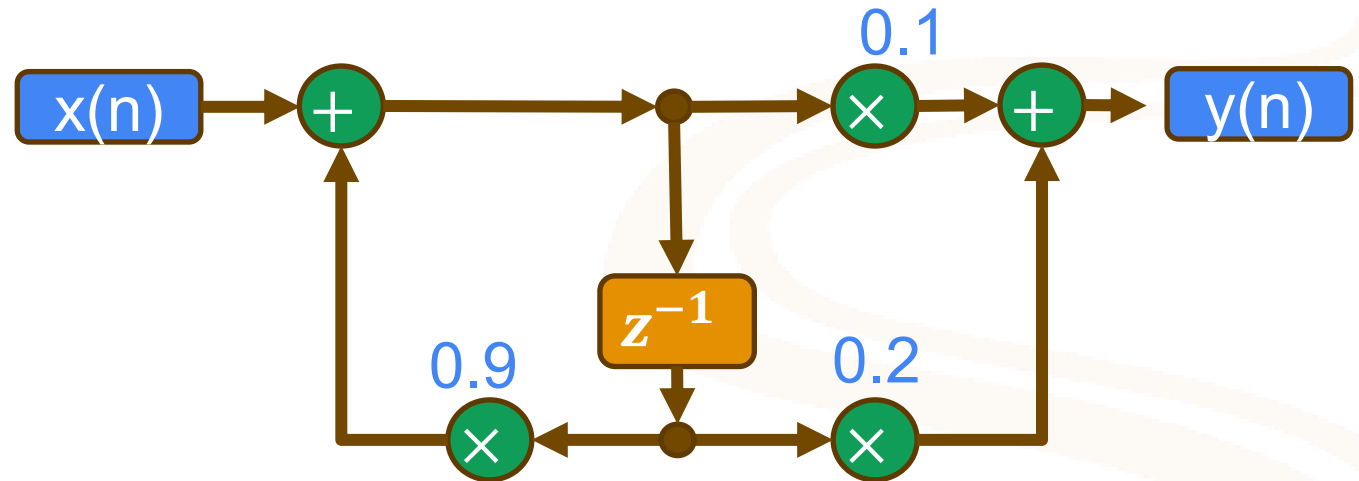
Invert all arrows

Mirror the diagram (optionally)

Proposed examples:

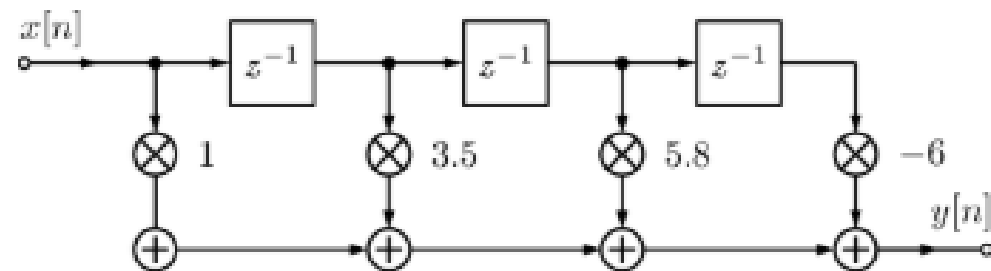
$$G_1(z) = \frac{0.01 - 0.12z^{-1} + 0.4z^{-3}}{1 - z^{-1} - z^{-2}}$$

$$G_2(z) = \frac{0.01 + 0.7z^{-4}}{1 - 0.9z^{-1} - 0.2z^{-3}}$$

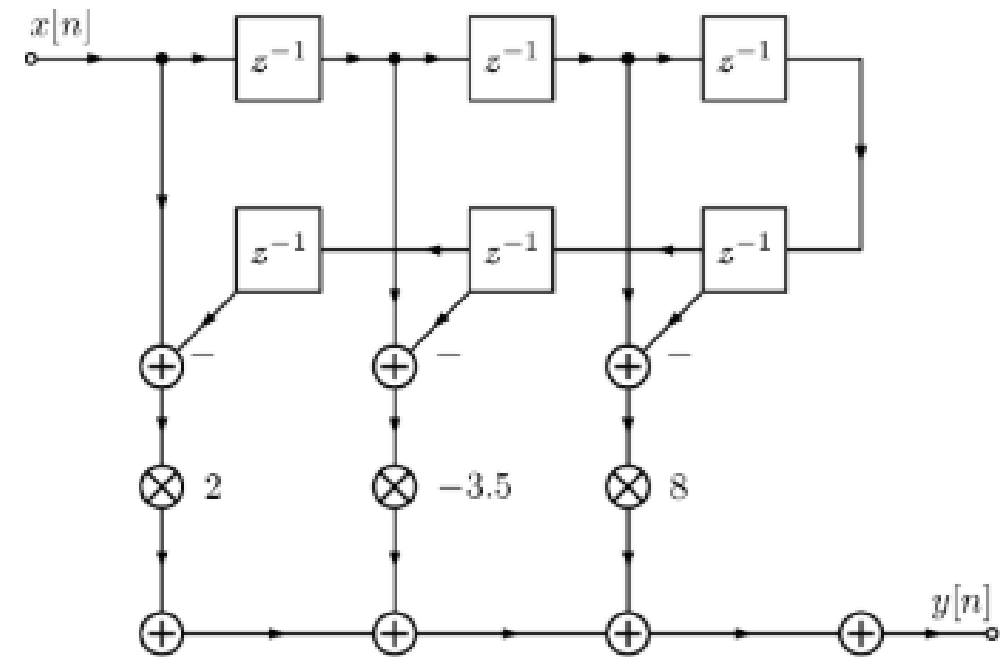


Zeros First
Poles Last

FIR realization



(a) *Direct form*



(b) *Symmetric*

Figure 3.59: *Two structures for FIR realizations.*



Thank you!

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